

# SIMATS ENGINEERING

## CO1 - ASSESSMENT TOOL 1

### Analytical Problem Solving

**COURSE NAME: DEEP LEARNING**

**COURSE CODE: MLA0408**

**FACULTY : Dr. D.SUDHAGAR**

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### COURSE OUTCOME COVERED

**CO 1:** Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

**CO1 Weightage: 15%**

**Assessment Tool Weightage: 35%**

**Duration: 30 minutes**

**Marks: 25**

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### Code of Conduct:

I GUZHALINITHA.K - 192425232 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



**Signature of the student:**

**QUESTIONS:05**

**Total Marks:25**

### Analytical Problem Solving

#### 1. Learning Algorithms (Optimization Behavior)

A machine learning model is trained using gradient descent on a convex loss function. Initially, the learning rate is set very high, and later it is gradually decreased.

#### Question:

Analyze how the choice of a high initial learning rate followed by decay affects convergence. Under what conditions can this strategy outperform a constant learning rate? Discuss possible risks and justify with reasoning.

## 2. Maximum Likelihood Estimation (MLE)

Suppose you are given a dataset generated from a Gaussian distribution with unknown mean  $\mu$  and known variance  $\sigma^2$

### Question:

Derive the Maximum Likelihood Estimator for  $\mu$ , and analytically explain how the estimate changes when:

- The dataset contains outliers
  - The sample size increases
- What does this imply about the robustness of MLE?

## 3. Building a Machine Learning Algorithm (Model Selection)

You are designing a classification algorithm for a dataset with 10,000 features but only 500 training samples.

### Question:

Analyze the challenges involved in training such a model. Propose a strategy (algorithm pipeline) to handle this situation and justify each step (e.g., feature selection, regularization, dimensionality reduction). Explain how your approach mitigates overfitting.

## 4. Neural Networks (Multilayer Perceptron - MLP)

Consider a Multilayer Perceptron with one hidden layer using sigmoid activation functions.

### Question:

Explain analytically why adding more hidden neurons increases the representational power of the network. Then, reason why this does not always lead to better performance. Support your answer using concepts like overfitting and generalization.

## 5. Backpropagation, SGD & Curse of Dimensionality

A deep neural network is trained using Stochastic Gradient Descent (SGD) on a very high-dimensional dataset.

### Question:

Analyze how the **curse of dimensionality** impacts:

- Gradient updates in backpropagation
- Convergence speed of SGD
- Model generalization

Propose at least two techniques to overcome these issues and justify them analytically.

## RUBRICS FOR CALCULATION:

| Criteria                         | Weightage | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    |
|----------------------------------|-----------|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20        | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts          | 30        | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |
| Problem-Solving Approach         | 20        | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |
| Accuracy of Solution             | 20        | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |
| Clarity                          | 10        | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |

### 1. Learning Algorithms (Optimization Behavior)

#### Answer:

Gradient Descent updates the model parameters using the learning rate ( $\eta$ ). A **high initial learning rate** allows the optimizer to make large steps toward the minimum of the loss function, resulting in faster learning during the initial stages. As training progresses, **learning rate decay** gradually reduces the step size, enabling fine-tuning near the optimum.

#### Effect on convergence:

- A high initial learning rate speeds up convergence by covering large distances quickly.
- Decreasing the learning rate later prevents overshooting and helps achieve a stable minimum.
- This combination usually converges faster than using a small constant learning rate.

#### When it outperforms a constant learning rate:

- When the loss function is smooth or convex.

- When the starting point is far from the optimal solution.
- In deep learning models where fast initial exploration and accurate final convergence are both required.

#### Possible risks:

- If the initial learning rate is too high, the algorithm may overshoot the minimum.
- The loss may oscillate or even diverge instead of converging.
- A very rapid decay may stop learning before reaching the optimal solution.

#### Justification:

A learning rate schedule combines the advantages of both large and small learning rates. It provides faster convergence initially and better stability in the final optimization stage, making it more efficient than a constant learning rate under suitable conditions.

## 2. Maximum Likelihood Estimation (MLE)

#### Answer:

For a Gaussian distribution with **unknown mean ( $\mu$ )** and **known variance ( $\sigma^2$ )**, the likelihood function is

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Taking the logarithm,

$$\log L(\mu) = \text{constant} - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Differentiating with respect to  $\mu$  and setting it equal to zero,

$$\frac{d}{d\mu} \log L(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0$$

Therefore,

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

Hence, the **MLE of the mean is the sample average**.

#### Effect of outliers:

- Outliers significantly affect the sample mean.
- A few extreme values can shift the estimated mean away from the true value.
- Therefore, MLE for the Gaussian mean is **not robust** to outliers.

#### Effect of increasing sample size:

- As the number of samples increases, random variations reduce.
- The estimated mean approaches the true population mean according to the Law of Large Numbers.
- The estimation variance decreases, improving accuracy.

**Implication about robustness:**

MLE provides efficient estimates for normally distributed data but is sensitive to outliers. Therefore, it is accurate for clean datasets but less robust when abnormal observations are present.

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### **3. Building a Machine Learning Algorithm (Model Selection)**

**Answer:**

The dataset contains **10,000 features** but only **500 training samples**, resulting in a high-dimensional problem.

**Challenges**

- High risk of overfitting.
- Curse of dimensionality.
- Increased computational complexity.
- Many irrelevant or redundant features.
- Poor generalization to unseen data.

**Proposed algorithm pipeline**

**Step 1: Data preprocessing**

- Handle missing values.
- Normalize or standardize features.

**Step 2: Feature selection**

- Remove irrelevant features using methods such as Chi-square, Mutual Information, or LASSO.
- This reduces noise and computational cost.

**Step 3: Dimensionality reduction**

- Apply Principal Component Analysis (PCA).
- Convert thousands of correlated features into a smaller number of informative components.

**Step 4: Regularization**

- Use L1 or L2 regularization.
- Penalize large weights and reduce model complexity.

**Step 5: Model training**

- Train a classifier such as Logistic Regression, SVM, or a simple Neural Network.

**Step 6: Cross-validation**

- Use k-fold cross-validation to evaluate model performance reliably.

**How overfitting is reduced**

- Feature selection removes unnecessary variables.
- PCA lowers dimensionality while preserving important information.
- Regularization prevents excessively large model weights.
- Cross-validation improves generalization and prevents memorizing the training data.

Thus, the pipeline produces a simpler, more accurate, and better-generalizing machine learning model.

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#### 4. Neural Networks (Multilayer Perceptron - MLP)

##### Answer:

A Multilayer Perceptron (MLP) with one hidden layer uses neurons to learn nonlinear relationships between inputs and outputs.

##### Why more hidden neurons increase representational power:

- Each hidden neuron learns different patterns or features.
- More neurons allow the network to represent more complex nonlinear functions.
- According to the Universal Approximation Theorem, an MLP with sufficient hidden neurons can approximate almost any continuous function.

##### Why more neurons do not always improve performance:

###### Overfitting

- Too many neurons may memorize the training data instead of learning general patterns.
- Training accuracy becomes very high, but testing accuracy decreases.

###### Generalization

- A good model should perform well on unseen data.
- Excessively large networks have higher variance and poorer generalization.

###### Computational cost

- More neurons increase training time, memory usage, and computational complexity.

###### Conclusion

Increasing hidden neurons improves the learning capacity of an MLP but must be balanced with regularization, sufficient training data, and validation techniques to achieve good generalization.

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#### 5. Backpropagation, SGD & Curse of Dimensionality

##### Answer:

The **curse of dimensionality** occurs when the number of features becomes extremely large, making learning difficult.

##### Impact on gradient updates during backpropagation

- High-dimensional parameter space produces noisy and unstable gradients.
- Weight updates become less effective.

- The optimizer requires more iterations to find an optimal solution.

### **Impact on convergence speed of SGD**

- SGD receives noisy gradient estimates.
- More dimensions increase optimization complexity.
- Training converges more slowly and may become unstable.

### **Impact on model generalization**

- High-dimensional data increases overfitting.
- Many features contribute little useful information.
- The model performs well on training data but poorly on unseen data.

### **Techniques to overcome these issues**

#### **1. Dimensionality Reduction (PCA or Autoencoder)**

- Reduces the number of input features.
- Removes redundant information.
- Improves convergence speed and reduces computational cost.

#### **2. Regularization (L1/L2 and Dropout)**

- Prevents excessively large weights.
- Reduces overfitting.
- Improves model generalization.

### **Additional techniques**

- Batch Normalization for stable gradient flow.
- Early Stopping to avoid overfitting.
- Feature Selection to retain only relevant features.